

Functional Features Included in the Solution

This document enumerates all functional features implemented in OptiCrop v1.0, categorised by MoSCoW priority and development status.

Feature Count Summary

Category	Planned	Implemented	Deferred
Must Have	7	7	0
Should Have	4	4	0
Could Have	3	2	1
Won't Have	2	0	2
TOTAL	16	13	3

Must Have Features (All Implemented)

#	Feature	Implementation Detail	Status
1	Crop Recommendation Form	7-input HTML form (N,P,K,temp,humidity,pH,rainfall)	Done
2	Input Validation (server)	validate_inputs() in preprocess.py with FEATURE_RANGES check	Done
3	Input Validation (client)	validate.js real-time range check before form submission	Done
4	ML Prediction Engine	Random Forest loaded from model.pkl via predict_crop()	Done
5	Result Display Page	result.html renders crop name, confidence score, crop image	Done
6	Mobile-Responsive Layout	Bootstrap 5 grid; tested at 375px, 768px, 1280px viewports	Done
7	Error Handling	Flash messages for validation errors; graceful 404/500 pages	Done

Should Have Features (All Implemented)

#	Feature	Implementation Detail	Status
1	Confidence Score	model.predict_proba().max() shown as percentage on result page	Done

2	Top-3 Crop Alternatives	Top 3 probabilities from predict_proba() shown with crop names	Done
3	Crop Profile Card	crops.json with description, season, water needs per crop	Done
4	Input Tooltip Guide	Bootstrap Tooltip on each form label with typical range info	Done

Could Have & Won't Have Features

#	Feature	Status	Note
1	Prediction Log (CSV)	Done	Appended to logs/predictions.csv on each POST /predict
2	About Page	Done	about.html with project info, team credits, dataset reference
3	Print / PDF Export of Result	Deferred	Scheduled for v1.2 requires wkhtmltopdf or Playwright
4	Hindi/Marathi language toggle	Won't Have v1	Scheduled for v1.1 requires Flask-Babel
5	WhatsApp bot integration	Won't Have v1	Scheduled for v2.1 requires Twilio API